ESP32 IoT LED Control using Blynk Cloud

Abstract

This project demonstrates remote control of an LED connected to an ESP32 using the Blynk IoT platform. A virtual switch in the Blynk dashboard sends commands to the ESP32 over Wi-Fi. The ESP32 receives the command and turns the LED ON or OFF accordingly.

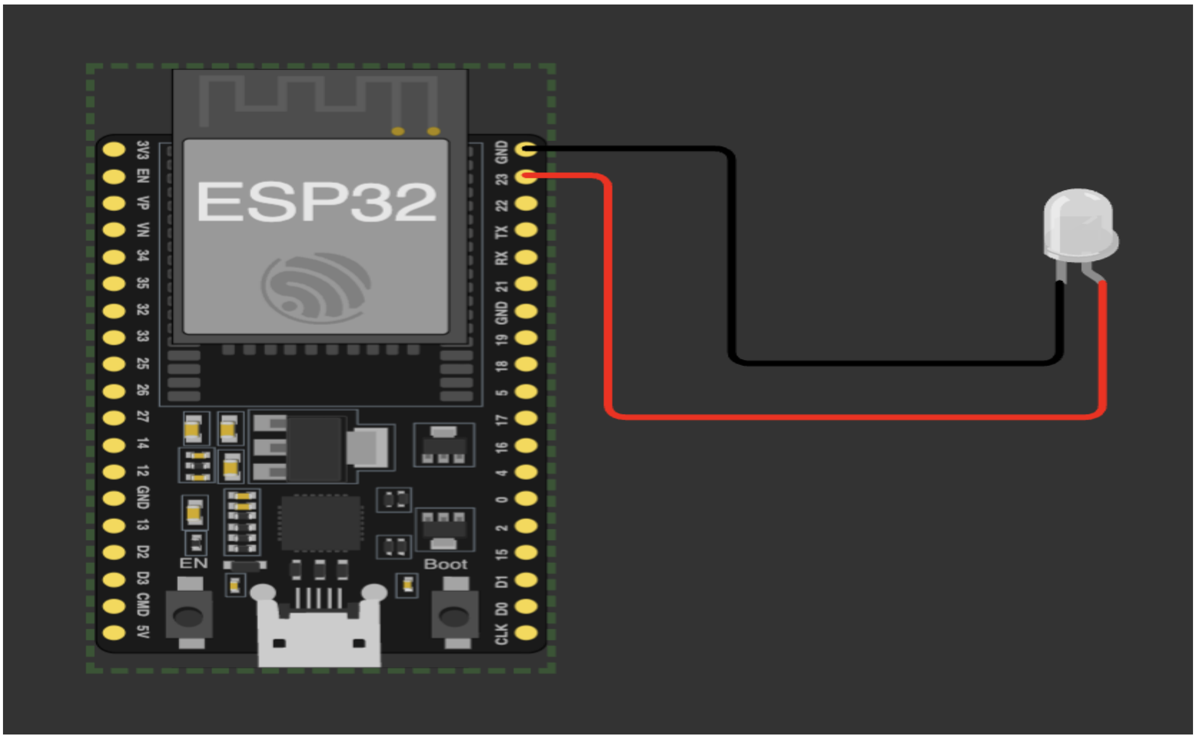
Components Required

• ESP32 Development Board
• LED
• Jumper Wires
• Wi-Fi Connection
• Blynk Cloud Platform

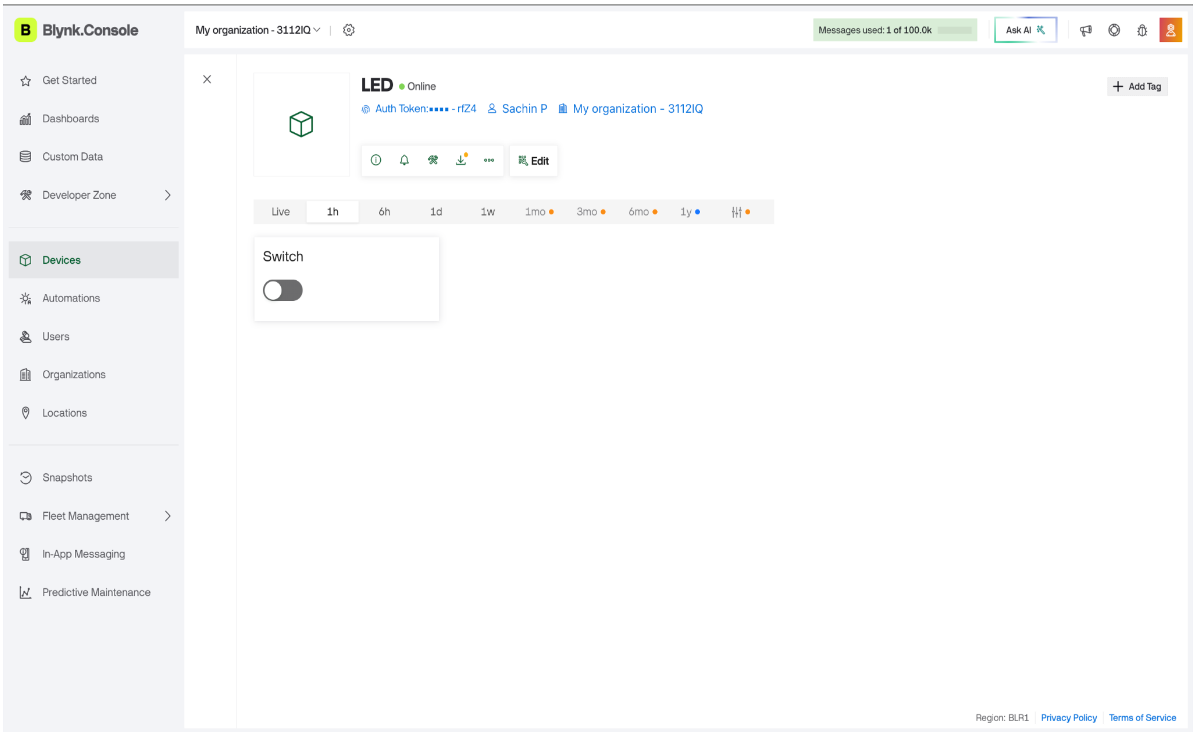
Circuit Description

The LED anode is connected to GPIO23 of the ESP32 and the cathode is connected to GND. The ESP32 connects to the Blynk Cloud using Wi-Fi credentials and an authentication token.

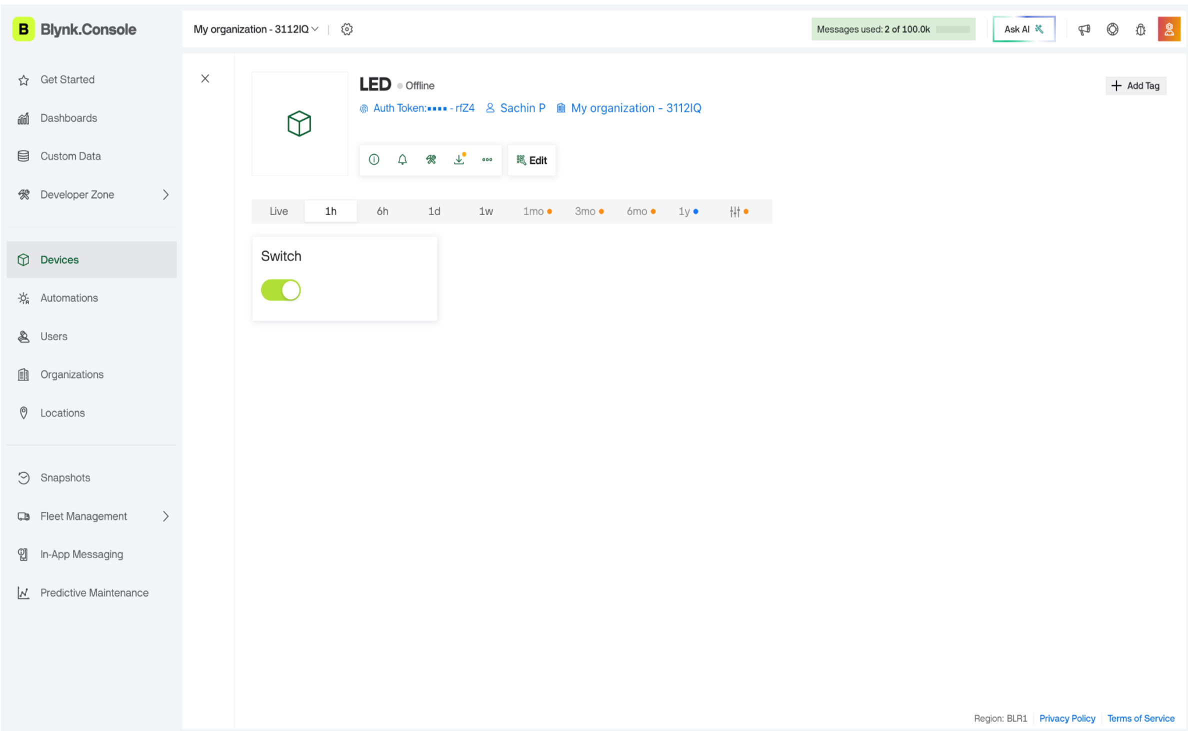
Circuit Diagram



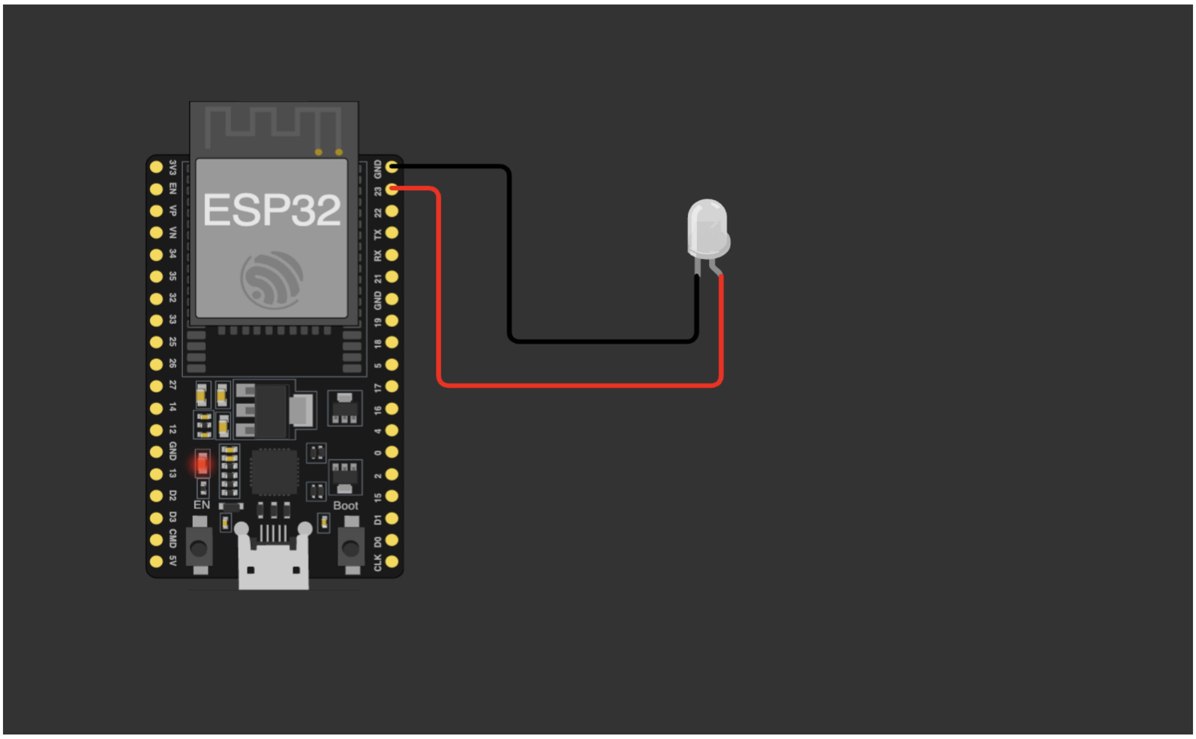
Blynk Dashboard - Switch OFF



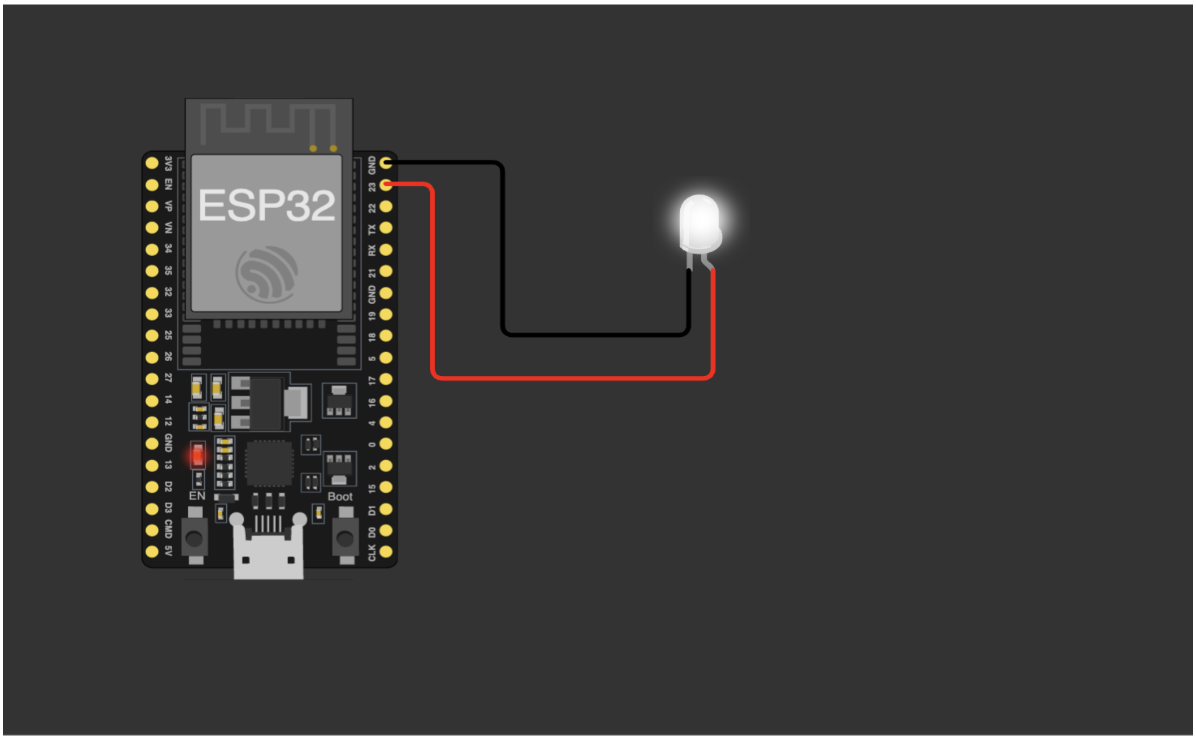
Blynk Dashboard - Switch ON



Output - LED OFF



Output - LED ON



Source Code

```
#define BLYNK_TEMPLATE_ID "TMPL3M97C-tIl"
#define BLYNK_TEMPLATE_NAME "LED"
#define BLYNK_AUTH_TOKEN "H-V5egntc86t2hQepfmMa2UOS-trrfZ4"
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

#define pin_led 23
#define WIFI_AP "Wokwi-GUEST"
#define WIFI_PASS ""

BLYNK_WRITE(V0)
{
  int pinValue = param.asInt();
  digitalWrite(pin_led, pinValue);
}

void setup() {
  // Initialize serial and Wi-Fi
  Serial.begin(115200);
  Serial.println("Hello, ESP32!");

  // Set LED pin as output
  pinMode(pin_led, OUTPUT);

  // Initialize Blynk with the authentication token and Wi-Fi credentials
  Blynk.begin(BLYNK_AUTH_TOKEN, WIFI_AP, WIFI_PASS);
}

void loop() {
  // Run Blynk repeatedly
  Blynk.run();
}
```

Working Principle

1. ESP32 connects to Wi-Fi and Blynk Cloud.
2. The user toggles the virtual switch in the Blynk dashboard.
3. Blynk sends the command to virtual pin V0.
4. The ESP32 receives the value through the BLYNK_WRITE(V0) callback.
5. GPIO23 is set HIGH or LOW, turning the LED ON or OFF.

Conclusion

The project successfully demonstrates IoT-based remote LED control using ESP32 and the Blynk platform. It provides a foundation for home automation and remote monitoring applications.